

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)	(i)		$F = ma$ or accept in words	1			1		
		(ii)	I	Total mass of train = $2.3 \times 10^5 \times 2$ (1) = $[4.6 \times 10^5]$ [kg] $= 4.6 \times 10^5 \times 0.7$ $= 3.22 \times 10^5$ (1) [N] accept 322×10^3 [N] Award maximum of 1 mark for $2.3 \times 10^5 \times 0.7 = 1.61 \times 10^5$ [N] accept 161×10^3 [N] Award 1 mark for answer of 3.22 or 1.61		2		2	2	
			II	Substitution: $0.7 = \frac{55.8}{t}$ (1) Rearrangement: time = 79.7 (1) [s] accept 80 [s]	1	1		2	2	
		(iii)		$2.3 \times 10^5 \times 2 = 4.6 \times 10^5$ (1) which is approximately 4.4×10^5 so this part of claim correct (1) Both trains have same emergency deceleration / 1.2 m/s^2 and so will take the same time to stop / this part of the claim is incorrect (1) Alternative: $\frac{4.4 \times 10^5}{2.3 \times 10^5} = 1.9$ (1) which is approximately 2 so this part of claim correct (1) Both trains have same emergency deceleration / 1.2 m/s^2 and so will take the same time to stop / this part of the claim is incorrect (1)			3	3	1	